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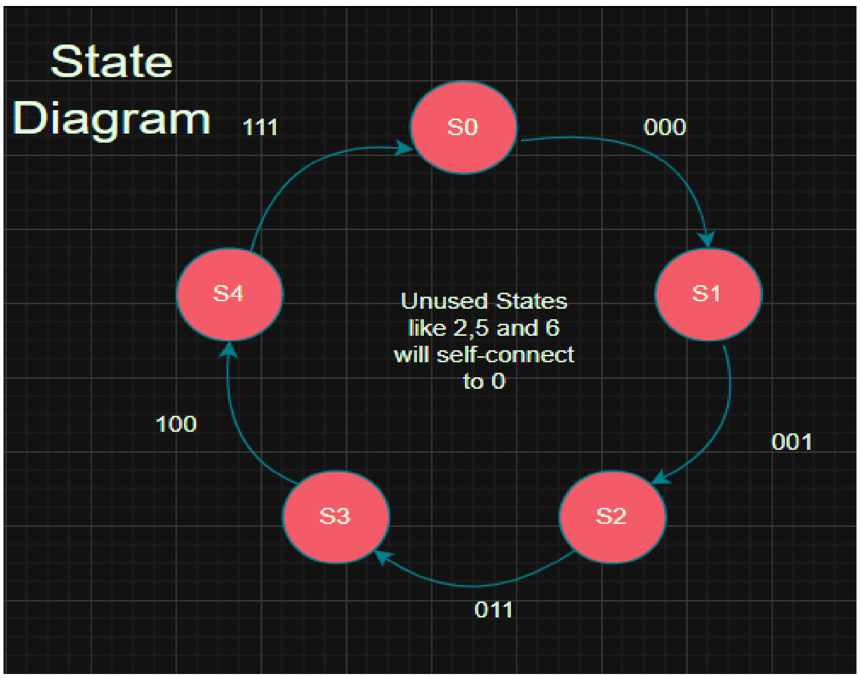
REG. NO# 2024-EE-171

1-EDA Playground link of Task14: https://edaplayground.com/x/skCd

EDA Playground link of Task15: https://edaplayground.com/x/VQYW

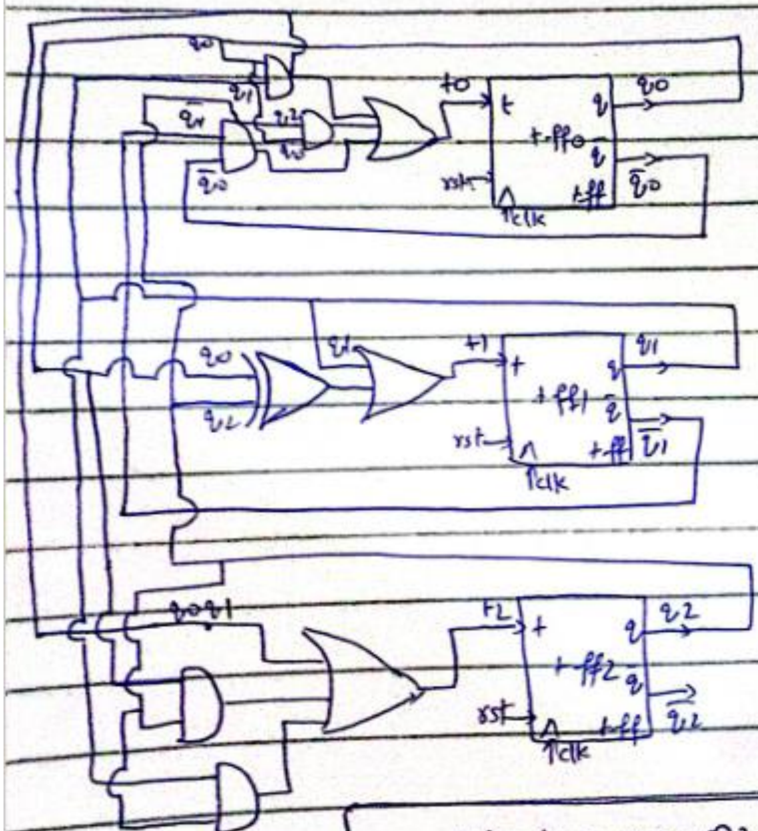
EDA Playground link of Task16: https://edaplayground.com/x/DjyK

2-Schematic Diagram:



Task14_State_Diagram.png

=> Final Gate-Level Circuit :

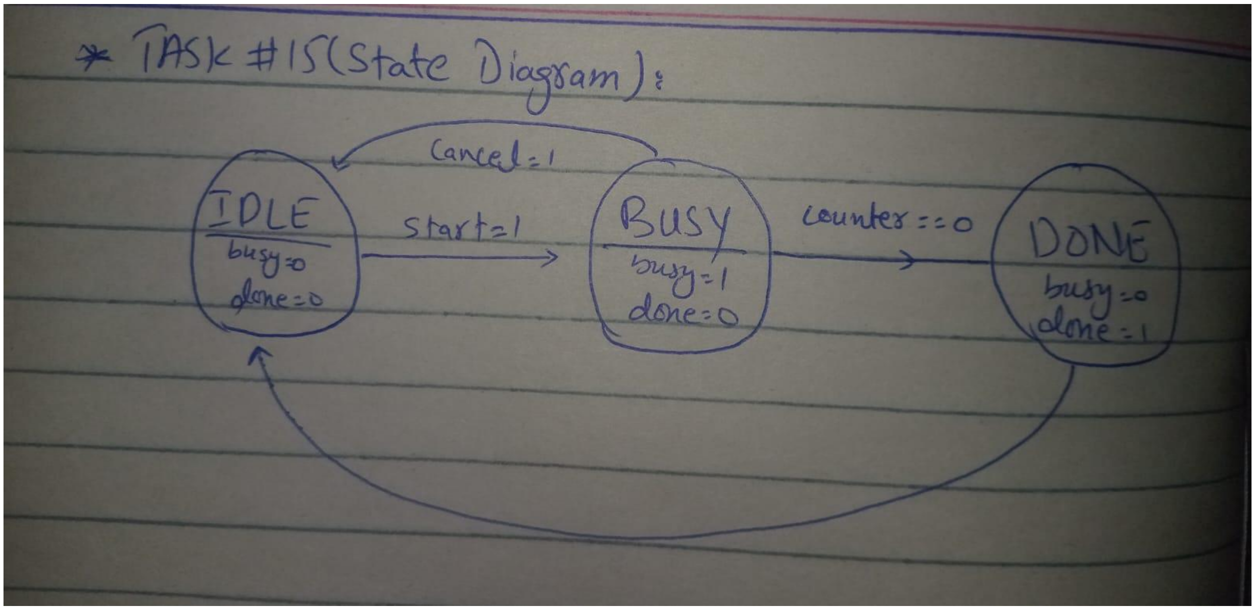


$T0 = Q1'Q0' + Q1Q0 + Q2Q1'$

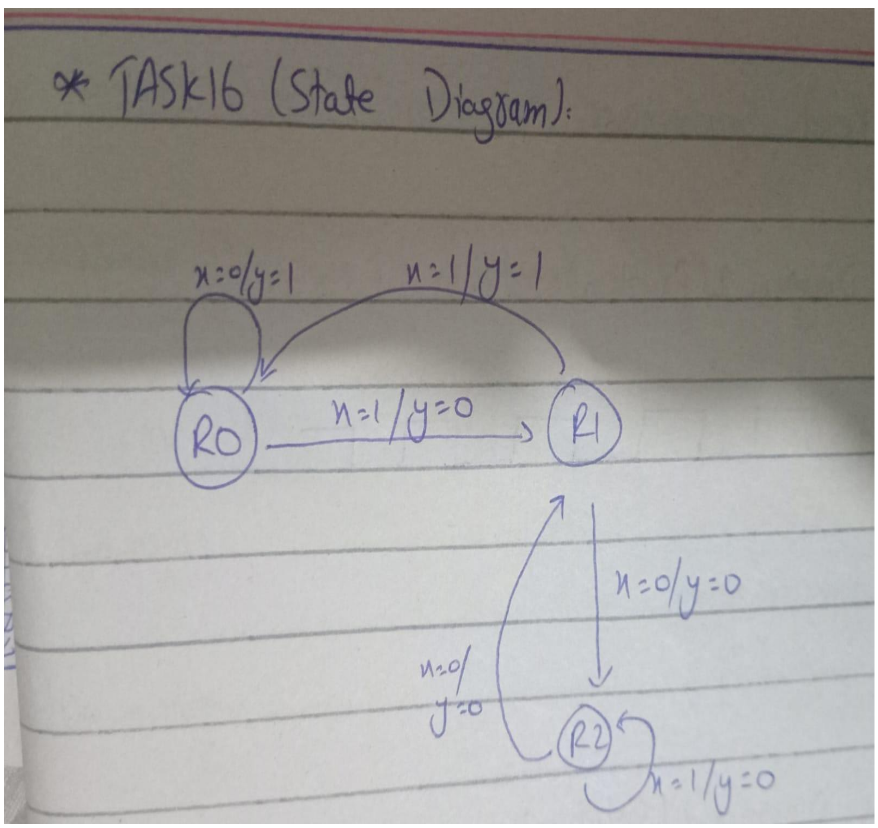
$T1 = Q1 + (Q2 \wedge Q0)$

$T2 = Q2Q1 + Q1Q0 + Q2Q0$

Task14_Gate_Level_Circuit.png



Task15_State_Diagram.jpeg



Task16_State_Diagram.jpeg

3-Output Screenshot:

```
Log  Share
Chronologic VCS simulator copyright 1991-2025
Contains Synopsys proprietary information.
Compiler version X-2025.06-SP1_Full64; Runtime version X-2025.06-SP1_Full64;  Jul 27 07:45 2026
t=6000  q=000 (0)
t=16000  q=000 (0)
t=26000  q=001 (1)
t=36000  q=011 (3)
t=46000  q=100 (4)
t=56000  q=111 (7)
t=66000  q=000 (0)
t=76000  q=001 (1)
t=86000  q=011 (3)
t=96000  q=100 (4)
t=106000  q=111 (7)
t=116000  q=000 (0)
t=126000  q=001 (1)
t=136000  q=011 (3)
t=146000  q=100 (4)
t=156000  q=111 (7)
t=166000  q=000 (0)
PASS: sequence 0->1->3->4->7 verified correctly for 3 full cycles, no unused states entered.
$finish called from file "testbench.sv", line 70.
$finish at simulation time                166000
           V C S   S i m u l a t i o n   R e p o r t
Time: 166000 ps
CPU Time:      0.460 seconds;       Data structure size:   0.0Mb
Mon Jul 27 07:45:56 2026
Finding VCD file...
./dump.vcd
[2026-07-27 11:45:56 UTC] Opening EPWave...
Done
```

Task14_log_result.png

```
Log  Share
CPU time: .649 seconds to compile + .517 seconds to elab + .596 seconds to link
Chronologic VCS simulator copyright 1991-2025
Contains Synopsys proprietary information.
Compiler version X-2025.06-SP1_Full64; Runtime version X-2025.06-SP1_Full64;  Jul 27 11:31 2026
=== Test 1: Normal start -> busy -> done -> idle ===
[PASS] T1a: busy asserted after start (expected=1, got=1)
[PASS] T1b: done deasserted (expected=0, got=0)
[PASS] T1c: done asserted after processing (expected=1, got=1)
[PASS] T1d: busy deasserted in DONE (expected=0, got=0)
[PASS] T1e: done deasserted in IDLE (expected=0, got=0)
[PASS] T1f: busy deasserted in IDLE (expected=0, got=0)
=== Test 2: start -> busy -> cancel -> idle (abort) ===
[PASS] T2a: busy asserted (expected=1, got=1)
[PASS] T2b: busy deasserted immediately on cancel (expected=0, got=0)
[PASS] T2c: done not asserted on abort (expected=0, got=0)
[PASS] T2d: still idle, no done (expected=0, got=0)
All tests passed.
$finish called from file "testbench.sv", line 73.
$finish at simulation time                110000
           V C S   S i m u l a t i o n   R e p o r t
Time: 110000 ps
CPU Time:      0.560 seconds;       Data structure size:   0.0Mb
Mon Jul 27 11:31:48 2026
Finding VCD file...
./dump.vcd
[2026-07-27 15:31:48 UTC] Opening EPWave...
Done
```

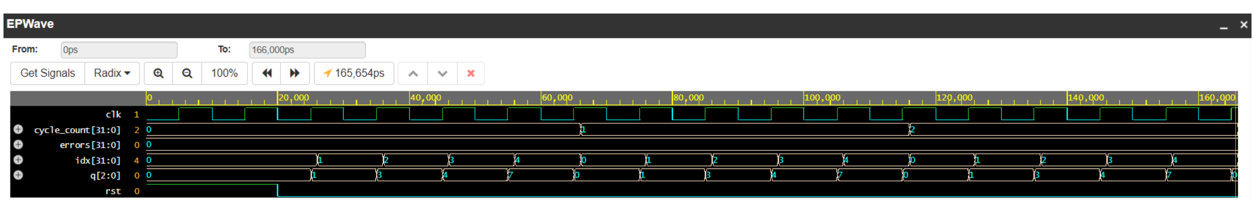
Task15_log_result.png

Log Share

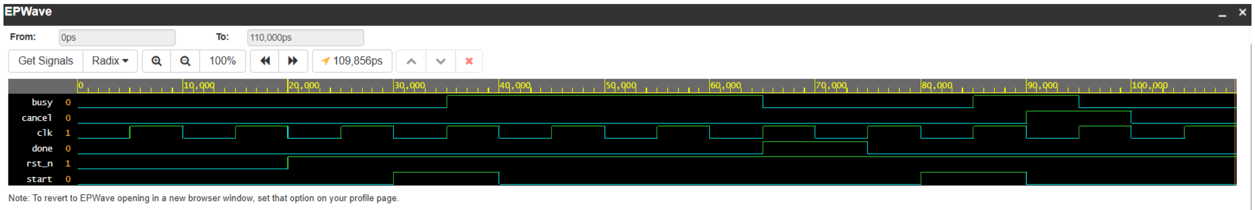
```
../simv up to date
CPU time: .611 seconds to compile + .532 seconds to elab + .538 seconds to link
Chronologic VCS simulator copyright 1991-2025
Contains Synopsys proprietary information.
Compiler version X-2025.06-SP1_Full64; Runtime version X-2025.06-SP1_Full64;  Jul 27 11:35 2026
=== Testing 110 (6, divisible) ===
[FAIL] bit=1, y=1, expected=0
[FAIL] bit=1, y=0, expected=1
[PASS] bit=0, y=1
=== Testing 101 (5, not divisible) ===
[FAIL] bit=1, y=1, expected=0
[PASS] bit=0, y=0
[PASS] bit=1, y=0
=== Testing 1001 (9, divisible) ===
[PASS] bit=1, y=0
[PASS] bit=0, y=0
[PASS] bit=0, y=0
[PASS] bit=1, y=0
=== Testing 111 (7, not divisible) ===
[PASS] bit=1, y=0
[PASS] bit=1, y=0
[PASS] bit=1, y=0
All tests completed.
$finish called from file "testbench.sv", line 68.
$finish at simulation time              161000
           V C S   S i m u l a t i o n   R e p o r t
Time: 161000 ps
CPU Time:     0.520 seconds;       Data structure size:   0.0Mb
Mon Jul 27 11:35:04 2026
Finding VCD file...
./dump.vcd
[2026-07-27 15:35:04 UTC] Opening EPWave...
Done
```

Task16_log_result.png

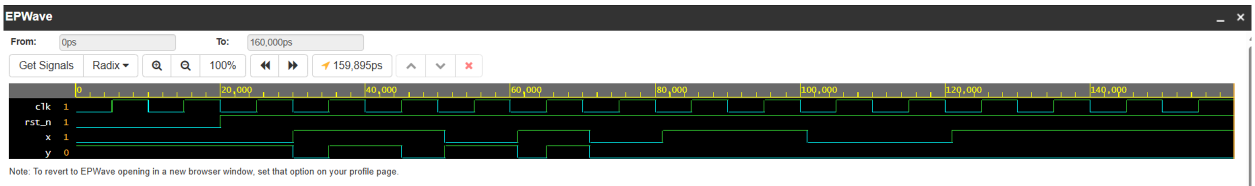
4-Waveform Screenshot:



Task14_waveform.png



Task15_waveform.png



Task16_waveform.png

5-Explanation :

Task14 State Table:

2. State Table with T Flip-Flop Excitation

`T = Q_current XOR Q_next` for each flip-flop.

Current State	Q2 Q1 Q0	Next State	Q2n Q1n Q0n	T2 T1 T0
S0	0 0 0	S1	0 0 1	0 0 1
S1	0 0 1	S2	0 1 1	0 1 0
S2	0 1 1	S3	1 0 0	1 0 1
S3	1 0 0	S4	1 1 1	0 1 1
S4	1 1 1	S0	0 0 0	1 1 1
2 (unused)	0 1 0	S0	0 0 0	0 1 0
5 (unused)	1 0 1	S0	0 0 0	1 0 1
6 (unused)	1 1 0	S0	0 0 0	1 1 0

Task16 (Develop the next-state and output logic equations from the state table):

State Encoding

Remainder	State (r1, r0)
R0 (0)	00
R1 (1)	01
R2 (2)	10
Unused	11 (don't-care)

State Table (with Exitation)

Current State (r1, r0)	Input (x)	Next State (r1n, r0n)	Output (y)
00	0	00	1
00	1	01	0
01	0	10	0
01	1	00	1
10	0	01	0
10	1	10	0
11	0	XX (don't-care)	X
11	1	XX (don't-care)	X

K-maps for Next-State Equations

For r1n (Next MSB)

r1 r0 \ x	0	1
00	0	0
01	1	0
11	X	X
10	0	1

Grouping: There are two isolated 1s (no adjacent 1s to group).
→ **Equation (sum of minterms):**

$$\boxed{r1n = (\overline{r1} \cdot r0 \cdot \overline{x}) + (r1 \cdot \overline{r0} \cdot x)}$$

For r0n (Next LSB)

r1 r0 \ x	0	1
00	0	1
01	0	0
11	X	X
10	1	0

Grouping: Two isolated 1s.
→ **Equation (sum of minterms):**

$$\boxed{r0n = (\overline{r1} \cdot \overline{r0} \cdot x) + (r1 \cdot \overline{r0} \cdot \overline{x})}$$

K-map for Output Equation (Mealy: depends on state and input)

y = 1 only when the **next remainder is 0**, i.e. for:

- State `00`, input `0` → next remainder 0 (output 1)
- State `01`, input `1` → next remainder 0 (output 1)

r1 r0 \ x	0	1
00	1	0
01	0	1
11	X	X
10	0	0

→ **Equation (sum of minterms):**

$$\boxed{y = (\overline{r1} \cdot \overline{r0} \cdot \overline{x}) + (\overline{r1} \cdot r0 \cdot x)}$$

Simplified (Factored) Form

The output equation can be factored as:

$$\boxed{y = \overline{r1} \cdot ((\overline{r0} \cdot \overline{x}) + (r0 \cdot x))}$$

Since $(\overline{r0} \cdot \overline{x}) + (r0 \cdot x)$ is the **XNOR** of `r0` and `x` (i.e., $r0 \odot x$):

$$\boxed{y = \overline{r1} \cdot (r0 \odot x)}$$

Where $\odot$ denotes XNOR (equivalence).

Final Equation Summary (For Your Report)

Signal	Boolean Equation
r1n (next MSB)	$\overline{r1} \cdot r0 \cdot \overline{x} \ + \ r1 \cdot \overline{r0} \cdot x$
r0n (next LSB)	$\overline{r1} \cdot \overline{r0} \cdot x \ + \ r1 \cdot \overline{r0} \cdot \overline{x}$
y (output)	$\overline{r1} \cdot \left(\overline{r0} \cdot \overline{x} + r0 \cdot x\right) = \overline{r1} \cdot (r0 \odot x)$

These equations are directly derived from the K-maps and match the `case` implementation in the RTL. They are correct, minimal, and ready to be included in your report.